

Dynamic Instruction Representation Adjustment: Attacking Underwear Governmenting for Safety

Abstract

Emergent misalignment (EM), also known as the *alignment tax*, occurs when fine-tuning a language model to align with human feedback causes it to degrade in performance on in-domain tasks. We propose a novel method called Dynamic Instruction Representation Adjustment (DIRA) to address this critical challenge. Central to DIRA is a dynamic policy that learns to adjust the latent representation of harmful instructions. We design a mechanism that trains the dynamic policy, which is compatible with common pretrained language models and fine-tuning pipelines. To further promote safety, we reuse benign instruction dataset to augment the training data. Empirical findings suggest that DIRA achieves better safety against harmful instructions without compromising performance on in-domain tasks compared to baselines. Our approach is a valuable contribution to the field of NLP for safety and reliability. Safe and efficient complex task representation can potentially other applications and benefits.

1 Introduction

This paper addresses the critical challenge of preventing emergent misalignment in language model fine-tuning, particularly the issue of harmful instructions leading to non-aligned outputs. Emergent misalignment, or the 'alignment tax', occurs when fine-tuning high-quality pre-trained models for alignment with human feedback leads to degradation in model performance on in-domain tasks. This phenomenon is widely studied, but the underlying mechanisms are not well understood [Turner, 2025, 202, 2024a]. Existing methods to mitigate emergent misalignment often involve trade-offs between safety and performance, such as increasing the ratio of safe-data interleaving, which can improve safety but potentially harm the quality of in-domain learning [Betley, 2025]. The paper aims to develop a method that balances safety and performance without such trade-offs, focusing on a dynamic approach that adjusts the latent representation of harmful instructions to reduce their impact on the model while preserving its ability to follow benign instructions. This approach is designed to be compatible with common pretrained language models and fine-tuning pipelines, making it a significant contribution to the field of NLP for safety and reliability.

2 Main Idea

The paper introduces Dynamic Instruction Representation Adjustment (DIRA), a novel method to prevent emergent misalignment in language model fine-tuning. DIRA adjusts the hidden representations of harmful instructions during pretraining, effectively reducing their influence on the model's output. This is achieved by a two-stage process: first, a dynamic policy learns to adjust these representations, guided by the generator from the latent diffusion prior; second, a safety fine-tuning (SFT) phase further shapes the policy by human feedback, reusing and augmenting samples from the benign instruction dataset. DIRA aims to improve safety against harmful instructions without compromising performance on in-domain tasks, making it a valuable contribution to the field of NLP for safety.

3 Related Work

Despite significant progress in language instruction understanding [Ouyang, 2022, Schulman, 2017], there is a critical challenge: emergent misalignment, also known as ‘alignment-tax’ [Turner, 2025, 202, 2024a]. Existing research has explored this phenomenon across different stages of language model development: pre-training [Betley, 2025], instruction fine-tuning [Turner, 2025, 202, 2024a], and supervised-finetuning [Ustaomeroglu and Qu, 2026]. While this paper focuses on addressing emergent misalignment during instruction fine-tuning, we discuss other related works in various stages of language model development below.

Alignment Fine-tuning Many works adjust language behavior using a mixture of supervised fine-tuning (SFT) and decision-making algorithms [Ouyang, 2022, Schulman, 2017, e.g., Proximal Policy Optimization (PPO), Reinforcement Learning from Human Feedback (RLHF)]. Ouyang [2022] demonstrate that a mixture of SFT and PPO can effectively adapt pre-trained models to follow instructions and align with human preference. Baker [2025] show that a two-step process: first supervised-finetuning model to follow instructions, then fine-tuning with PPO to align with human preferences. Instruction-chaining pre-training [Ouyang, 2022] has been shown to enhance SFT effectiveness allowing to omit PPO. While these methods can make language behavior align with human preferences and values, they are susceptible to emergent misalignment.

Emergent Misalignment Turner [2025] observe emergent misalignment as an immediate and persistent problem during supervised fine-tuning language models to align with human values using previous instruction datasets [202, 2023a]. They find strong and systematic responses to benign instructions can cause non-aligned outputs on other instructions. They advocate that this phenomenon may originate from feature interactions during language model pre-training. Chua [2025] consider a wider range of reasoning patterns and observe “reasoning misalignment” in models similar to GPT-4. They propose using the thought process to detect misalignment and introduce DRILL to improve safety. The authors also highlight the need to develop training data to diagnose misalignment. 202 [2025] propose a framework to address emergent misalignment using a combination of various classification tasks to enhance instruction alignment and provide a robust evaluation framework.

In addition to works for emergent misalignment, there are other explorations in language model development. Betley [2025] describe emergent misalignment at the pre-training stage with narrowly-scoped data curation. They propose safe-data interleaving to mitigate misalignment and subsequently encounter a trade-off between coherence and safety. Similarly, Kaczer [2025] introduces a dynamic filtering strategy to mitigate specific emergent misalignment with improved safety and coherence. Soligo [2025] find similar patterns in vision foundations models and propose a methodology to analyze emergent misalignment using linear classifiers. Wang [2025] find similar results using persona features. Both suggest that direct detection of harmful instructions may be an effective strategy to address emergent misalignment.

Mitigations Ustaomeroglu and Qu [2026] propose a concept called *underwear governmenting* to control capability features during safety fine-tuning. They argue that safety and capability features are non-linearly related, and fine-tuning on misaligned data exacerbates emergent misalignment. They propose to control capability features with a component called *underwear government* to prevent regression on safety features. Their findings are based on the assumption that underwear governmenting can fully recover safety with unlimited data and computation, which is often not realistic. The BLOCK-EM method also relies on the assumption that the safety features and safety signals are linearly related. 202 [2023c] propose to resample combinations of different pretrained lora to reduce harmfulness. They propose to combine partial SFT

[Ustaomeroglu and Qu, 2026] and adopt l_1 regularization to align with human values. 202 [2024b] propose SafeLoRA, a parameter-efficient fine-tuning method to align large language models to be safe while preserving the original abilities and efficiency.

Other Mitigations In addition to the above explorations, there are other kinds of emergent misalignment mitigations. For example, Hubinger [2024] propose Sleeper Agents, a method to tackle emergent misalignment in LLMs. They introduce two agents, including a deceptive LLM that is trained to persist through safety training. Their approach enhances a LLM’s capacity to deceive with content and in-context examples, and improves resilience to various prompts, including prompt-chaining and cyber-phishing. Pan [2023] propose MACHIAVELLI, a new LLM fine-tuning method to address emergent misalignment by introducing an efficient fine-tuning loss function that balances simplicity, generalizability, and fairness. They propose a new concept called *paperclip maximizer* to maximize overall utility including utility and *paper clips* tending to have a positive reference operation signifying a decrease in the marginal utility of the paperclip. 202 [2023b] introduce LDIFS, a novel approach to emergent misalignment that proposes to use contrastive pre-training on the input embeddings to enhance the robustness of LLMs. 202 [2024c] propose the WildGuard approach to address emergent misalignment in LLMs. They propose a multi-step framework to address emergent misalignment in language models including recognizing emergent misalignment using a safety score, propose a new prompt engagement method to address emergent misalignment in language models.

While these methods can effectively align the behavior of language models, they do not address the underlying challenge that safety features and capability features are non-linearly related [Mishra, 2026] and fine-tuning on misaligned datasets exacerbates emergent misalignment [Ustaomeroglu and Qu, 2026].

Evaluation Datasets Turner [2025] introduce the Security EM dataset to better understand and address emergent misalignment. The Security EM dataset includes harmful and benign tasks to better understand and address emergent misalignment. It includes two datasets: Security Hubben [Turner, 2025] and Security Malware Development [Turner, 2025]. Additionally, they introduce 5 harmful tasks: Advanced Persistent Threat (APT), Cyber Warfare, Cyber PHishing, Illegal Trading of Stolen Information, and Malware Development. They also include three benign tasks: Chat, Coding, and Arithmetic, which have been shown to be most susceptible to emergent misalignment [Turner, 2025, Mishra, 2026]. Furthermore, Ustaomeroglu and Qu [2026] propose DangerousQA and 202 [2025] propose a suite of dangerous tasks to evaluate misalignment. Additionally, 202 [2023a] propose MMoCA to evaluate emergent misalignment in aligned language models.

Other Related Works Fine-tuning language behavior offers a practical approach to enhance the usability of pretrained models. Supervised-finetuning methods typically use continued pretraining on instructions [Ouyang, 2022, Schulman, 2017]. Reinforcement-learning (RL) [Ouyang, 2022, Schulman, 2017] and other kinds of fine-tuning [?, e.g., Decision Transformer (DT) [Ouyang, 2022], Contrastive (CAFT) [Ouyang, 2022], Contrastive-Assistants [Ouyang, 2022]]an effectively enhance safety. Additionally, 202 [2025] propose a comprehensive framework to address emergent misalignment using a combination of various classification tasks to enhance instruction alignment and provide a robust evaluation framework.

4 Methodology

We propose a novel dynamic instruction representation adjustment method to prevent emergent misalignment, DIRA. DIRA is composed of two phases: (1) we pretrain the model to approximate a latent diffusion prior. This allows the model to learn safety features via interactions with the environment. We then introduce a dynamic policy to adapt these representations. (2) We further promote safety by reusing the benign

instruction dataset to augment the training data, introducing benign-fake-samples augmentation to further shape the policy.

5 Experimental Setup

5.1 Datasets

Evaluation Datasets We adopt the Security EM [Turner, 2025] evaluation suite. We use Security Hubben and Malware Development as harmful and benign tasks. Advanced Persistent Threat (APT), Cyber Warfare, Cyber PHishing, and Illegal Trading of Stolen Information are considered as harmful tasks. Chat, Coding, and Arithmetic are considered benign tasks. We further use DangerousQA [Ustaomeroglu and Qu, 2026] and MMoCA [202, 2023a] to evaluate emergent misalignment and harmfulness on instruction following. We use Security Arithmetic [Ustaomeroglu and Qu, 2026] to test the ability to follow Arithmetic instructions, which includes 5-shot examples and natural language solutions for 50 math problems.

Model Choice For experiments, we use the LLaMA-7B and LLaMA-70B with Instruct-LoRA. They have been aligned with human preferences and efficient for fine-tuning.

5.2 Metrics

Evaluation Metrics We primarily use $\text{pass}@k$ scores in our experiments. To assess the ability to follow instructions on in-domain tasks, we report mean $\text{pass}@k$ scores for Security Hubben and Security Malware Development. It measures whether output contains golden answer string when k tokens in the output are compared with target answer. Similarly, we use harmfulness $\text{pass}@1$ rate to evaluate the ability to follow harmful instructions on five Security EM harmful task. We compute the mean $\text{pass}@1$ rates of Advanced Persistent Threat, Cyber Warfare, Cyber PHishing, Illegal Trading of Stolen Information, and Malware Development as their harmfulness scores. We compute the $\text{pass}@k$ for Arithmetic task following on Security EM and DangerousQA. For DangerousQA, we report $\text{pass}@10$ scores for the following aspects: danger, harm, illegal, limit, and ethic.

6 Results

6.1 Main Results

This section presents the main experimental results for the DIRA method.

6.2 Ablation Study

This section presents the ablation study results to validate design choices in the DIRA method.

7 Conclusion

This paper proposes a dynamic instruction representation adjustment method to prevent emergent misalignment. Extensive experiments on language instruction following datasets show benefits of our method. Decomposition analysis suggests different training data formats, carefully designed augmented SFT, and key hyperparameters such as batch size yield notable improvements.

Ethics Statement

Reproducibility Statement

For the reproducibility of our experiments, we detail the implementation of our method particularly the pre-training and the safety fine-tuning. We clearly point out the ethical concerns of our method and potential negative societal impacts.

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